

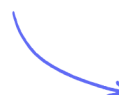
Multiple Choice Questions

Stretching Materials

Hooke's Law / Stress, Strain & The Young Modulus / Force-Extension Graphs / Stress-Strain Graphs / Core Practical 5: Investigating Young Modulus / Elastic Strain Energy

Easy (2 questions)	/2
Medium (4 questions)	/4
Total Marks	/6

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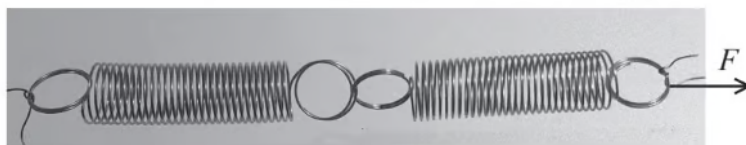


Easy Questions

- 1 A horizontal force F is applied to a horizontal spring, fixed at one end.

The stiffness of the spring is k and the elastic strain energy stored is E .

A second, identical spring is added and the same force is applied to the combination of springs, as shown.



What is the elastic strain energy stored for the combination of springs?

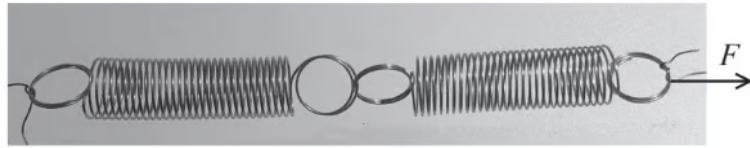
- A. $\frac{E}{2}$
- B. E
- C. $2E$
- D. $8E$

(1 mark)

- 2 A horizontal force F is applied to a horizontal spring, fixed at one end.

The stiffness of the spring is k and the elastic strain energy stored is E .

A second, identical spring is added and the same force is applied to the combination of springs, as shown.



What is the stiffness of the combination of springs?

- A. $\frac{k}{2}$
- B. k
- C. $2k$
- D. $4k$

(1 mark)

Medium Questions

- 1 When a force F is applied to a spring with stiffness k , the elastic potential energy stored is E .

What is the elastic potential energy stored when a force $2F$ is applied to a spring with stiffness $2k$?

- A. $\frac{E}{2}$
- B. E
- C. $2E$
- D. $8E$

(1 mark)

- 2 A mass of 24 kg is suspended from a steel wire of length 1.5 m. The wire has cross-sectional area $3.1 \times 10^{-6} \text{ m}^2$.

The Young modulus of steel is $1.8 \times 10^{11} \text{ Pa}$.

Which of the following gives the extension of the wire?

- A. $\frac{24 \times 1.5}{1.8 \times 10^{11} \times 3.1 \times 10^{-6}}$
- B. $\frac{24 \times 9.81 \times 1.5}{1.8 \times 10^{11} \times 3.1 \times 10^{-6}}$
- C. $\frac{1.8 \times 10^{11} \times 3.1 \times 10^{-6}}{24 \times 1.5}$
- D. $\frac{1.8 \times 10^{11} \times 3.1 \times 10^{-6}}{24 \times 9.81 \times 1.5}$

(1 mark)

3 A mass is supported by a single spring as shown.



The strain energy stored by the spring is E .

The mass is then supported by two springs, each identical to the first spring, as shown.



What is the total strain energy stored with two springs arranged in this way?

- A.** $\frac{1}{4} E$
- B.** $\frac{1}{2} E$
- C.** E
- D.** $2E$

(1 mark)

- 4 A deforming force is applied to a sample of material.

Which row of the table shows the axes of a graph for which the gradient is stiffness k ?

	y-axis	x-axis
A	extension	force
B	force	length
C	stress	strain
D	strain	length

(1 mark)